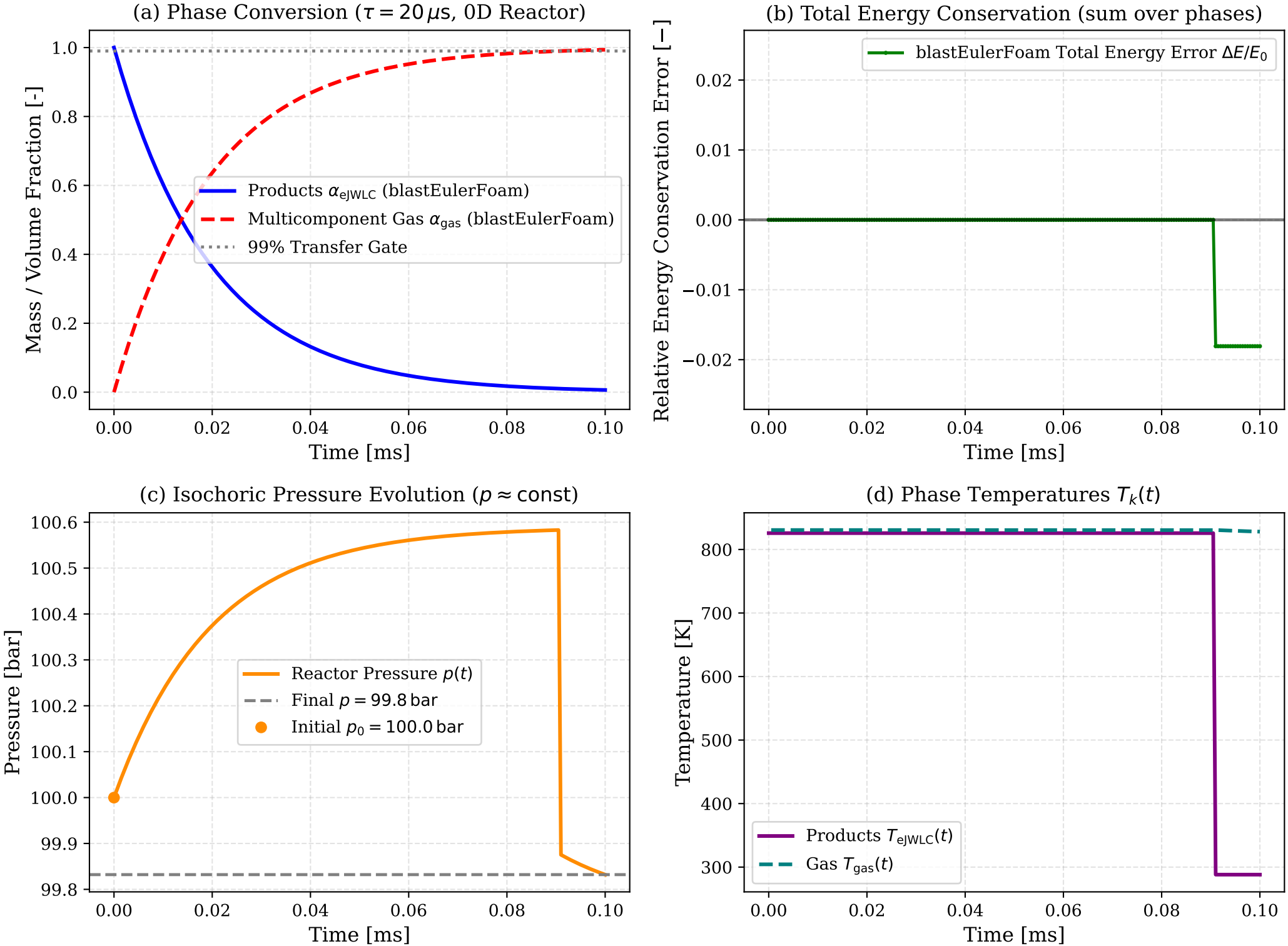
(a) Phase Conversion ($\tau = 20\,\mu$s, 0D Reactor)
Mass / Volume Fraction [-]
Time [ms]
Products α_{eJWLC} (blastEulerFoam)
Multicomponent Gas α_{gas} (blastEulerFoam)
99% Transfer Gate
(b) Total Energy Conservation (sum over phases)
Relative Energy Conservation Error [−]
Time [ms]
blastEulerFoam Total Energy Error $\Delta E/E_0$
(c) Isochoric Pressure Evolution ($p \approx$ const)
Pressure [bar]
Time [ms]
Reactor Pressure $p(t)$
Final $p = 99.8$ bar
Initial $p_0 = 100.0$ bar
(d) Phase Temperatures $T_k(t)$
Temperature [K]
Time [ms]
Products $T_{\mathrm{eJWLC}}(t)$
Gas $T_{\mathrm{gas}}(t)$